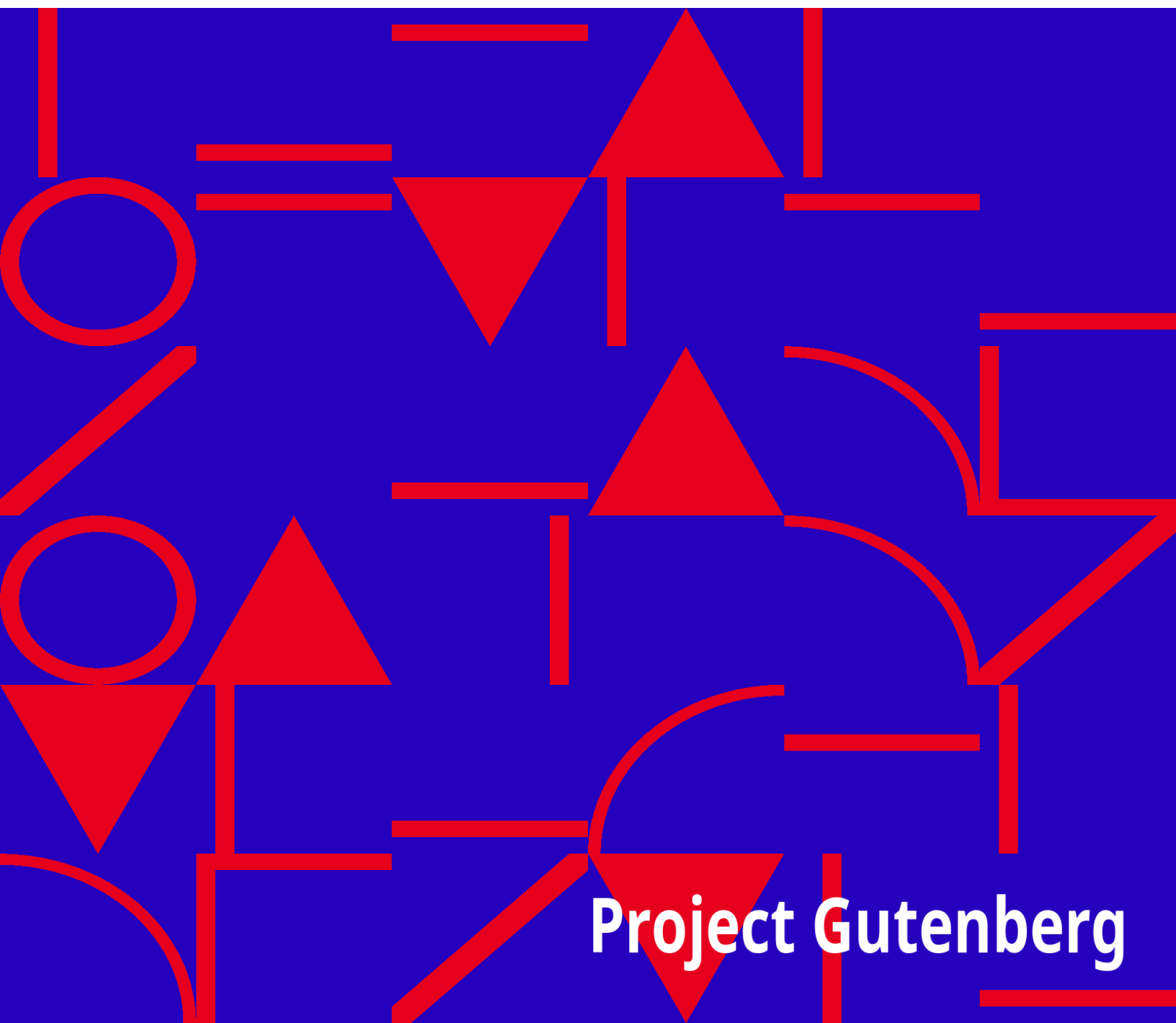


Great Western Railway Instructions

William Wylie Grierson



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Great Western Railway.

INSTRUCTIONS.

SUPERELEVATION OR “CANT.”

EXPANSION OF RAILS.

GAUGE AND CHECKING OF CURVES.

W. W. GRIERSON,
Chief Engineer.

PADDINGTON,
June, 1922.

SUPERELEVATION OR “CANT.”

To be given to the Outer Rail in Curves ranging from 6 to 200 Chains Radius.

Offset (Length of Cord 66 ft.)	Radius of Curve.	MINIMUM “CANT.”		MAXIMUM “CANT.”	
		Outer rail to be raised.	For a maximum speed of	Outer rail to be raised.	For a maximum speed of
Inches.	Chains.	Inches.	miles per hr.	Inches.	miles per hr.
16½	6	3¾	20	6	25
10	10	3½	25	5⅛	30
6⅝	15	3⅜	30	4⅝	35
5	20	3½	35	5¾	45
4	25	2¾	35	4⅝	45
3¼	30	3	40	4¾	50
2¾	35	2⅝	40	4	50
2½	40	2¼	40	5⅛	60
2¼	45	2½	45	4½	60
2	50	2¼	45	5⅝	70
1⅝	60	1⅞	45	4⅝	70
1⅜	70	1⅝	45	4	70

$1\frac{1}{4}$	80	$1\frac{3}{4}$	50	$3\frac{1}{2}$	70
$1\frac{1}{8}$	90	$1\frac{1}{2}$	50	$3\frac{1}{8}$	70
1	100	$1\frac{3}{8}$	50	$2\frac{3}{4}$	70
$\frac{7}{8}$	120	$1\frac{1}{8}$	50	$2\frac{3}{8}$	70
$\frac{3}{4}$	140	1	50	2	70
$\frac{5}{8}$	160	$\frac{7}{8}$	50	$1\frac{3}{4}$	70
$\frac{1}{2}$	200	$\frac{3}{4}$	50	$1\frac{3}{8}$	70

The amount of “cant” to be given to a curve is governed by the speed at which trains run over it, and in no case must the raising of the outer rail in Running Lines be less than is given in the column headed “Minimum Cant,” except where necessary to meet special cases, as at switches, crossings and junctions. In curves on falling gradients where high speeds are run, the outer rail must be kept well up, while on rising gradients a lesser “cant,” but within the limits of the latter, must be given. On gradients on single lines the outer rail of curves must, as a rule, be raised to suit the speed of descending trains.

The superelevation must be gradually attained at the rate of 1 inch in each 66 feet length as the curve is approached, maintained uniformly over the whole length of the curve, and run out after passing the curve in the same way.

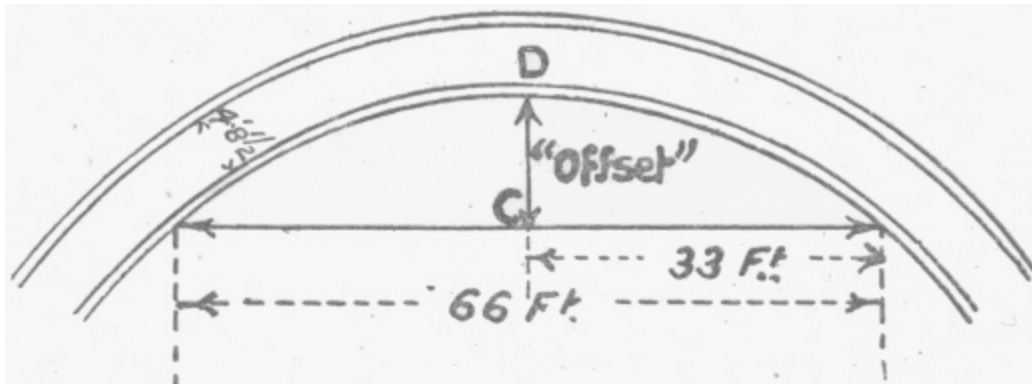
On reverse curves, or curves with very short pieces of straight between them, the “cant” must be made gradually on the curves commencing at the middle of the short piece of straight in the latter case.

EXPLANATION.

Stretch a cord measuring exactly a chain (66 feet) across the inner side of the curve, and measure the distance (C D) between the inner edge of the rail and the cord, as shewn in sketch below.

Look for this measurement in the column headed “Offset,” opposite to which, in the column headed “Radius of Curve,” will be found the radius of

the curve in chains; and in the columns headed “Minimum Cant” or “Maximum Cant” will be found how much the outer rail is to be raised above the inner one, according to the speed of trains.



EXAMPLE.

Suppose the distance between the inner edge of the rail and the cord at C D is found to be $2\frac{1}{2}$ inches, the column headed “Radius of Curve,” shews it to have a curve of 40 chains radius; and those headed “Minimum Cant” or “Maximum Cant” will shew that the outer rail must be raised $2\frac{1}{4}$ inches higher than the inner one for a speed of 40 miles an hour, or $5\frac{1}{8}$ inches for a speed of 60 miles an hour.

EXPANSION OF RAILS.

In laying the 45 ft., 44 ft. 6 in., 39 ft. 5 in., 32 ft., 29 ft. and 26 ft. lengths of rails, the following spaces must be left at the joints, according to the temperature for the expansion and contraction of the metal, viz.:—

Air Temperature. (Degrees Fahrenheit.)	NATURE OF WEATHER.	Length of Rail.	
		45' 0"	32' 0"
		44' 6"	29' 0"
		39' 5"	26' 0"
80	In hot Summer weather	$\frac{5}{32}"$	$\frac{3}{32}"$
60	In moderately cool weather	$\frac{1}{4}"$	$\frac{3}{16}"$
30	In cold Winter weather	$\frac{13}{32}"$	$\frac{9}{32}"$

For laying down new permanent way iron expansion gauges, equal in thickness to the respective spaces given, to be used.

Wooden gauges must on no account be allowed.

The expansion granges must, as far as possible, be allowed to remain in the joints until they are properly fished up, and the road slewed into its proper line and “topped.”

GAUGE AND CHECKING OF CURVES.

When laying new lines, or relaying in curves of 10 chains radius and above in plain line work, the gauge of the road must be laid to the standard width of 4 ft. 8½ in., but in curves of less than 10 chains radius, worked over by Passenger and Goods Trains, the gauge may be slightly eased according to the radius of the curve and other circumstances.

The check rail in plain road is to be second-hand bull-head rail, and all curves of Passenger lines of 10 chains radius, and under, must be continuously checked; the check rail in all cases to extend, where possible, for a length of 50 feet or thereabouts outside the commencing and terminating points of the curve to be checked.

Continuous check chairs for plain road, are of two patterns, one allowing a clearance of 1¾ in. to be used in curves of 10 chains radius, and the other allowing a clearance of 2 in. to be used in curves under 10 chains, both rails in each, case being level on top.

In sharp reverse or “S” curves in plain line work where a piece of straight line cannot be given, the junction of the two curves must be laid with check rails on both sides, each check to be laid to cover the other for a safe distance.

*** END OF THE PROJECT GUTENBERG EBOOK GREAT WESTERN
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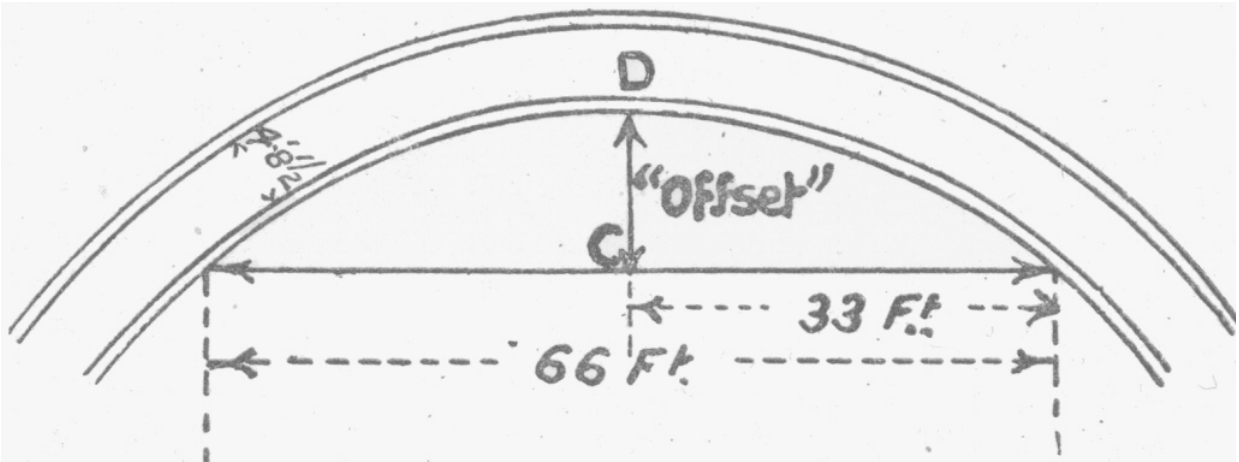
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